

Fedora Hardware validation QA framework

This qualification process started at Meta with the intention to be shared broadly through the Fedora Ready program.

The main premise of the Fedora hardware validation workflow shared here is to provide some guidance and tools for automated testing in order to assess a device usability, user experience and level of support for the Fedora operating system. The tests conducted are split into 2 categories:

1. Automated testing - This repository contains a bootable installer ISO based on the Fedora Everything netinstaller, a kickstart for automated provisioning, and some additional scripts for testing. These tests are a combination of known open-source stress tests and benchmarks.
2. Manual testing - This guide also provides documentation on some standard QA practices and tests that can be conducted manually, as well as documentation templates for completing a report which will mark a device as officially Fedora supported.

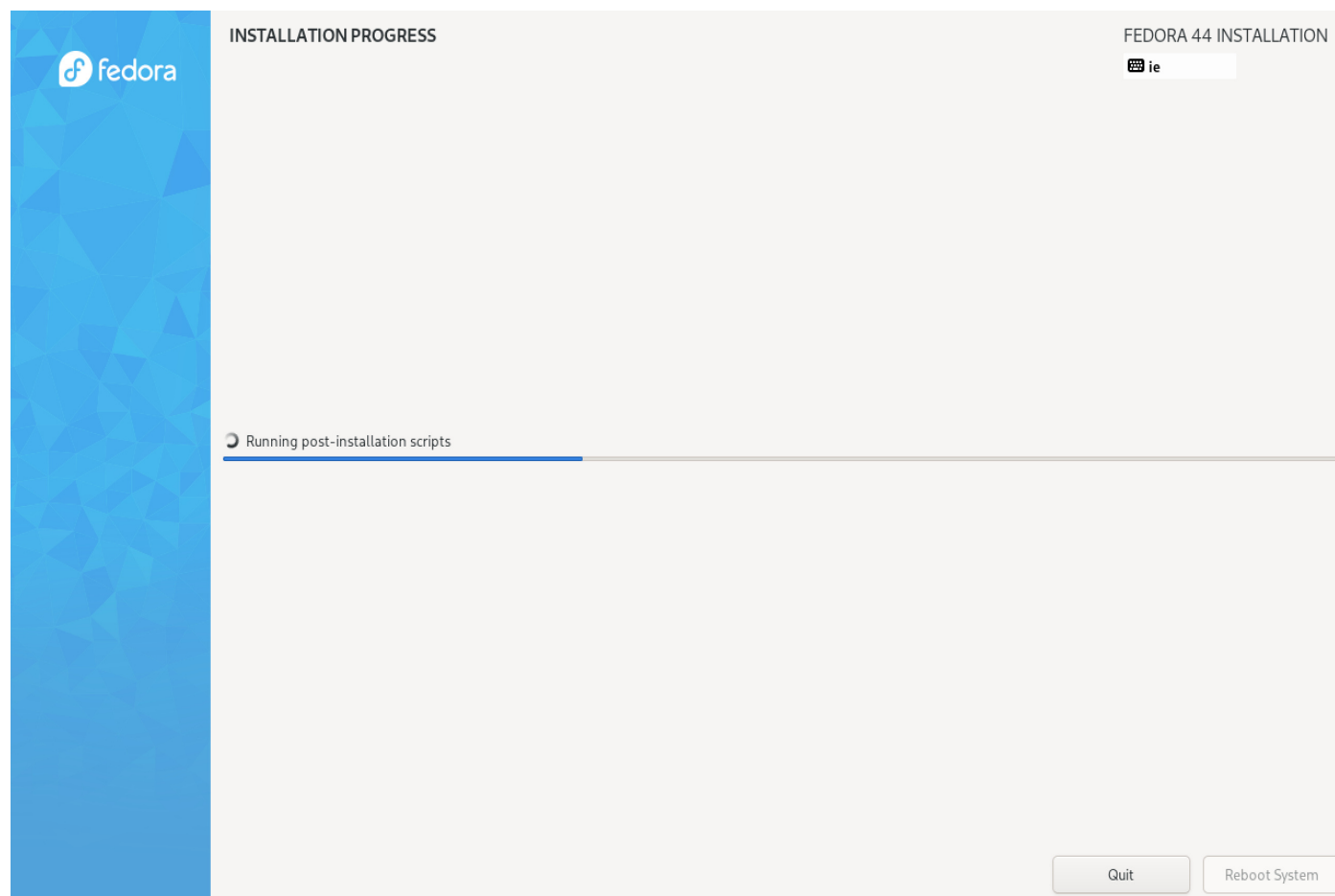
Automated testing with custom ISO

ISO Instalation Step by step

It is recommended to use a bootable USB drive with [Ventoy](#).

Also note that if you have secure boot enabled, automatic MOK key enrollment for the akmods key will be attempted. This is to ensure that self-signed kernel modules are loaded on boot, like for example on devices with NVidia GPUs.

1. Boot the `fedora-fantasy-hwqual.iso` from your USB drive
2. The installer will automatically proceed if a network connection is detected. If not, connect to your network manually when prompted.
3. Allow the installation to proceed. The `post_kickstart` script will execute during the "Running post-installation scripts" step (progress can be monitored under TTY1 - Press CTRL + ALT + F1)

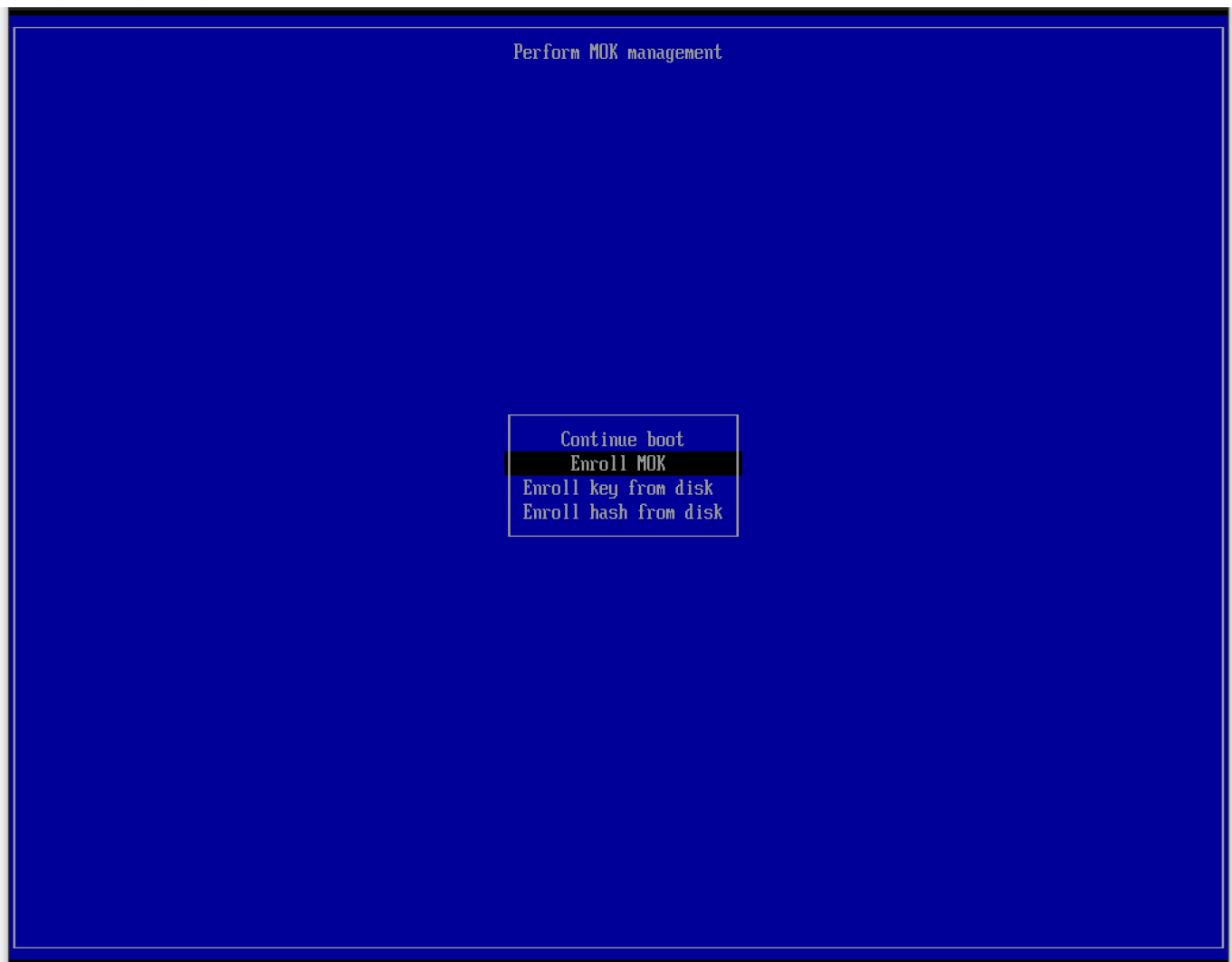


1. If secure boot is enabled you will be prompted to create a temporary password for the akmods MOK key. Once you set a password, press any key to resume the installation.

```
#####  
#  Secure Boot: MOK Key Enrollment      #  
#  You will be prompted to set a one-time password.  #  
#  Remember it - you'll need it on the next reboot.  #  
#####  
  
input password:  
input password again:  
  
MOK key queued for enrollment. On next boot, the MOK  
manager will ask you to confirm with the password above.  
  
Press any key to continue..._
```

NOTE: The password here is completely temporary and will only be used during MOK Key enrollment. Something simple like "1234" is perfectly acceptable.

1. If everything went well, the device will automatically reboot. On the next reboot, MOK Key enrollment will be automatically initiated.

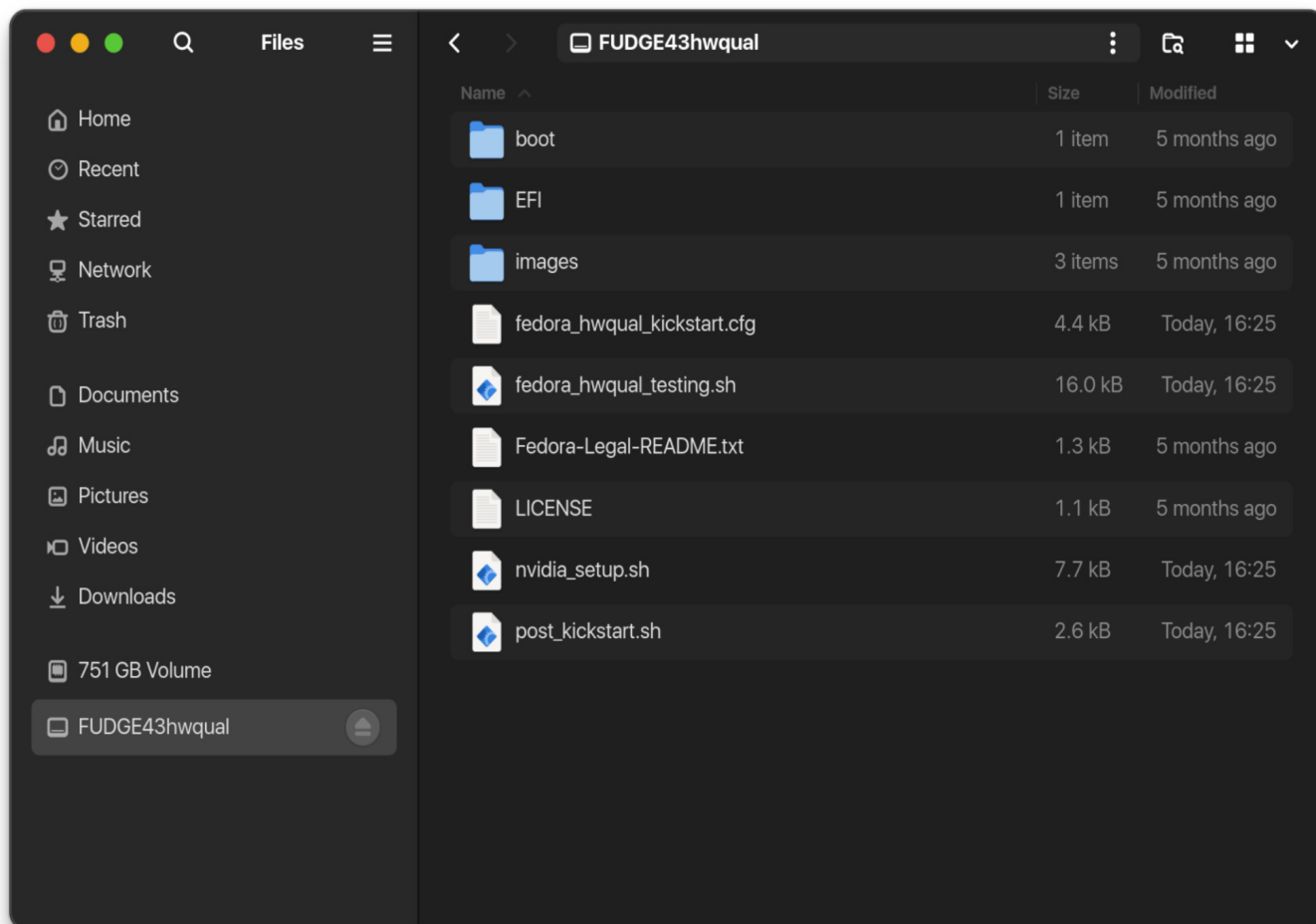


1. Proceed to enroll the akmods key to the BIOS following the prompts on the screen and by using the temp password you configured at the previous step.
2. Once you complete the MOK key enrollment, reboot the device.
3. Complete the Welcome to Fedora guide to create a user account etc.
4. The device is now ready to use for further testing.

ISO structure

The ISO is built on top of the Fedora Everything netinstaller ISO as found on public Fedora repositories here: <https://fedoraproject.org/misc/#everything>

We include in the ISO the following 4 files which can also be found on the root path of the ISO when mounted as well as this repository.



Phase 1 - Anaconda instalation

fedora_hwqual_kickstart.cfg

Kickstart is an answer file which can be provided to a netinstaller and automate the steps otherwise performed by user selections (for example, system language, desktop environment etc)

This kickstart roughly does the following:

- Defines the primary source repo to get official Fedora packages
- Answers some automation basics like language, region etc.
- Partitions a drive with a 2GB EFI partition, a 2GB EXT4 partition for /root and the rest of the drive as BTRFS
- Encrypts the drive with the simple passphrase "fedora"
- Installs the packages needed for @^workstation-product-environment
 - (this is the default workstation option when using the Fedora live Gnome ISO too)
- If all goes well, preps post_kickstart to start next.

Kickstart then goes to fedora_stable_post_kickstart.sh.

post_kickstart.sh

This is a script that is executed in the %post section of the anaconda kickstart flow which allows us to run certain steps additionally to a typical kickstart install.

Post kickstart does the following:

- Moves the scripts baked into the ISO to the /usr/local/bin/ system path
- Updates the kernel to the latest version as well as all system updates.
- Runs some additional package installs with dnf, like Google Chrome, benchmark tooling like phoronix-test-suite, glmark2 etc.
- Creates a randomly generated file used as a secondary LUKS decryption key so on reboot, the password prompt can be skipped

- Executes the next script bundled in the ISO to install Nvidia drivers if needed

nvidia_setup.sh

This script will handle Nvidia driver installation if Nvidia hardware is detected on the system. It can run under two different modes, rpmfusion and nvidia.

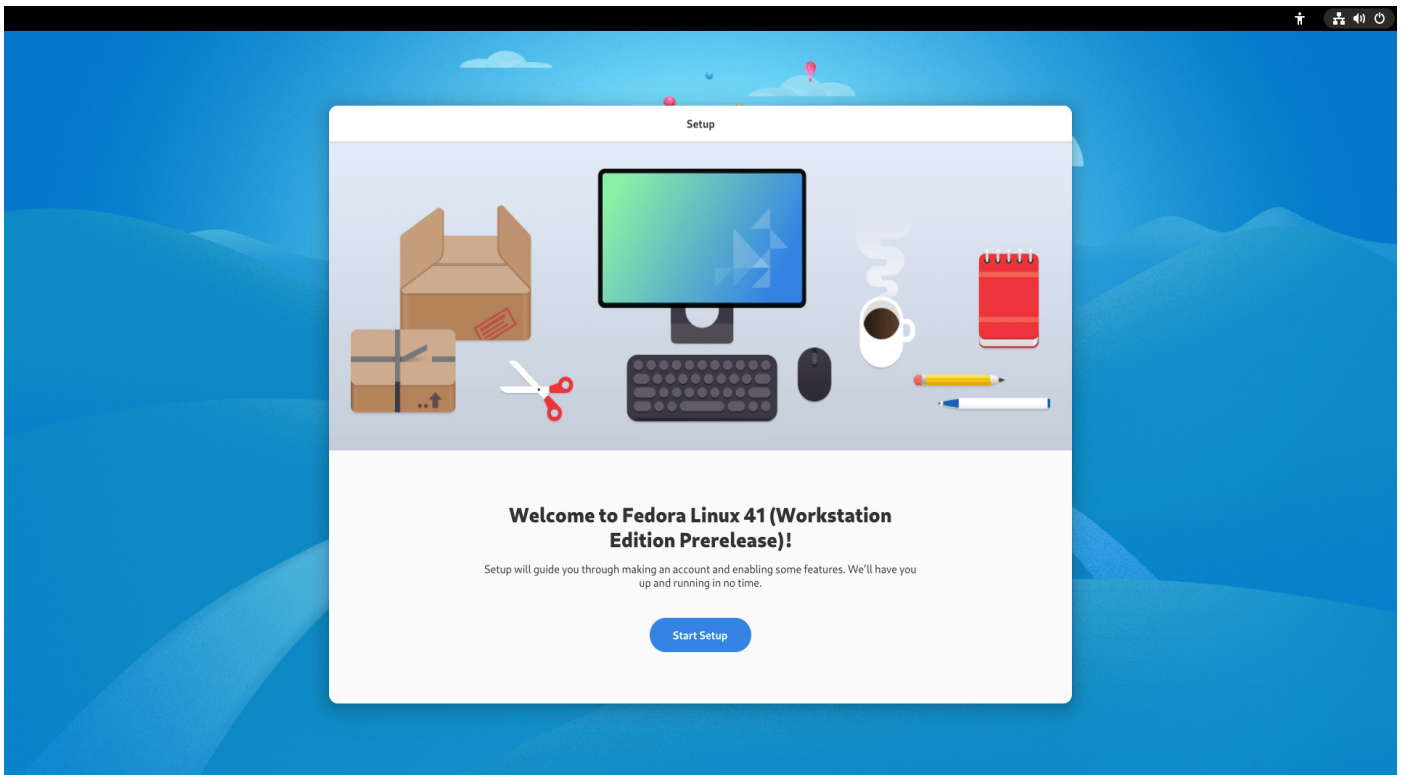
The first mode will install Nvidia drivers by enabling 3rd party rpmfusion repositories and use the driver provided by that repo and the open source kernel modules handled by akmod. This option is also the default during automated installer.

The second mode will install the official Nvidia driver from Nvidia's cuda repository, using dkms to manage the provided kernel module. This is an option we provide to our Linux users if they require cuda libraries for development and they can uninstall the rpmfusion drivers and re-execute this script manually to switch to the official drivers instead of the rpmfusion ones.

After that the Anaconda part of the installation will be complete and the device will restart. The output of all the above scripts will be written in the log file at `/root/ks-post.log`.

Phase 2 - Welcome to Fedora OOB manual setup

The device will automatically restart after the Anaconda environment has finished installing the OS and the Welcome to Fedora OOB screen will appear:



fedora_hwqual_testing.sh

This script is also added to the ISO but not executed automatically. It will be moved by the kickstart phase to `/usr/local/bin/` so it can be called from the terminal.

Benchmark script design (fedora_hwqual_testing.sh)

As mentioned above, the script is automatically added in `/usr/local/bin/` so it can be easily called from a terminal window with the command: `fedora_hwqual_testing.sh`

The script will ask for sudo permissions due to certain actions requiring elevated privileges (installing dependencies etc.)

It will then start running in a tmux environment. The primary steps will be displayed below the welcome art while the actual tests will be running in a split window on the side.

Script UI:

2. iozone benchmark
3. hdparm speedtest
4. fio speedtest through the phoronix-test-suite
5. dbench speedtest through the phoronix-test-suite

Operating System

- Basic operating system functions testing with osbench through the phoronix-test-suite

Final report

Once all tests are performed an HTML file report will be generated which will create a summary of the device specifications and the tests performed. The Device specification will be gathered with the tool inxi. The final report will look like this:

Hardware Stress Testing and Benchmarks Report

Start Time: Wed 25 Mar 2026 6:26:39

System Info		Test Results			
Category	Value	Category	Test	Sample Result	
System	System: Host: angelotheo-fedora-PW0MBZL0 Kernel: 6.18.16-200.fc43.x86_64 arch: x86_64 bits: 64 Console: pts/3 Distro: Fedora Linux 43 (Workstation Edition)	CPU Stress Test	stress	stress: info: [677046] successful run completed in 300s	
			cachebench	Average: 82775.417784 MB/s	
Machine	Machine: Type: Laptop System: LENOVO product: 21TES0BQ00 v: ThinkPad T1g Gen 8 serial: PW0MBZL0 Mobo: LENOVO model: 21TES0BQ00 v: SDK0T76530 WIN serial: W1HV5BE10BL Firmware: UEFI vendor: LENOVO v: N4EET21W (1.07) date: 12/29/2025	CPU Benchmark	coremark	Average: 336381.793335 Iterations/Sec	
			graphics-magick	Average: 64 Iterations Per Minute	
			epoch	Epoch3D Deck: Cone: 688.53 Average: 688.53 Seconds	
CPU	CPU: Info: 16-core model: Intel Core Ultra 7 265H bits: 64 type: MCP cache: L2: 28 MiB Speed (MHz): avg: 1583 min/max: 400/5300:4600:2500 cores: 1: 1583 2: 1583 3: 1583 4: 1583 5: 1583 6: 1583 7: 1583 8: 1583 9: 1583 10: 1583 11: 1583 12: 1583 13: 1583 14: 1583 15: 1583 16: 1583		deepspeech	Acceleration: CPU: 100.8183 Average: 100.8183 Seconds	
Graphics	Graphics: Device-1: Intel Arrow Lake-P [Arc Pro 130T/140T] driver: i915 v: kernel Device-2: NVIDIA GB206M [GeForce RTX 5070 Max-Q / Mobile] driver: nvidia v: 580.126.18 Device-3: Luxvisions Innotech Integrated Camera driver: hid-sensor-hub,usbhid,uvcvideo type: USB Display: unspecified server: X.Org v: 24.1.9 with: Xwayland v: 24.1.9 driver: gpu: i915 resolution: 3840x2400-120Hz API: OpenGL v: 4.6.0 vendor: nvidia v: 580.126.18 renderer: NVIDIA GeForce RTX 5070 Laptop GPU/PCIe/SSE2 API: EGL Message: EGL data requires eglinfo. Check --recommends. Info: Tools: api: glxinfo gpu: nvidia-settings,nvidia-smi x11: xdriinfo, xdpinfo, xprop, xrandr	GPU Stress Test	furmark	Average: 28.097122 FPS	
		GPU Benchmark	glmark2-wayland	glmark2 Score: 2321	
			unigine-super	Average: 44.2 Frames Per Second	
Memory	Memory: System RAM: total: 64 GiB available: 62.21 GiB used: 8.01 GiB (12.9%) Array-1: capacity: N/A slots: 8 note: check modules: 8 EC: None Device-1: Controller0-ChannelA-DIMM0 type: LPDDR5 size: 8 GiB speed: spec: 8533 MT/s actual: 7467 MT/s Device-2: Controller0-ChannelB-DIMM0 type: LPDDR5 size: 8 GiB speed: spec: 8533 MT/s actual: 7467 MT/s Device-3: Controller0-ChannelC-DIMM0 type: LPDDR5 size: 8 GiB speed: spec: 8533 MT/s actual: 7467 MT/s Device-4: Controller0-ChannelD-DIMM0 type: LPDDR5 size: 8 GiB speed: spec: 8533 MT/s actual: 7467 MT/s Device-5: Controller1-ChannelA-DIMM0 type: LPDDR5 size: 8 GiB speed: spec: 8533 MT/s actual: 7467 MT/s Device-6: Controller1-ChannelB-DIMM0 type: LPDDR5 size: 8 GiB speed: spec: 8533 MT/s actual: 7467 MT/s Device-7: Controller1-ChannelC-DIMM0 type: LPDDR5 size: 8 GiB speed: spec: 8533 MT/s actual: 7467 MT/s Device-8: Controller1-ChannelD-DIMM0 type: LPDDR5 size: 8 GiB speed: spec: 8533 MT/s actual: 7467 MT/s	RAM Stress Test	blender	Blend File: BMW27 - Compute: CPU-Only: 111.07 Blend File: Junkshop - Compute: CPU-Only: 149.02 Blend File: Classroom - Compute: CPU-Only: 285.85 Blend File: Fishy Cat - Compute: CPU-Only: 163.82 Blend File: Barbershop - Compute: CPU-Only: 1172.33 Blend File: Pabellon Barcelona - Compute: CPU-Only: 379.16	
			memtester	Done.	
			sysbench memory	Total operations: 58081278 (5807477.33 per second) 56720.00 MiB transferred (5671.36 MiB/sec)	
Audio	Audio: Device-1: Intel Arrow Lake cAVS driver: sof-audio-pci-intel-mtl API: ALSA v: k6.18.16-200.fc43.x86_64 status: kernel-api	RAM Benchmark	tinymembench	Average: 58410.8 MB/s Average: 58410.8 MB/s	
			ramspeed	Average: 32133.87 MB/s	
Network	Network: Device-1: Intel Arrow Lake CNVi WiFi driver: iwlwifi	Drive Stress Test	sysbench fileio	reads/s: 399.65 writes/s: 266.44 fsyncs/s: 852.81 read, MiB/s: 6.24 written, MiB/s: 4.16 total time: 300.1083s	
Bluetooth	Bluetooth: Device-1: Intel driver: btusb type: USB Report: btmgmt ID: hci0 state: up address: C8:95:CE:A5:8D:7B bt-v: 5.4		iozone	Avg throughput per process = 1191302.50 kB/sec	
Drives	Drives: Local Storage: total: 953.87 GiB used: 100.01 GiB (10.5%) ID-1: /dev/nvme0n1 vendor: Samsung model: MZVLCT0HFLU-00BLL size: 953.87 GiB		hdparm	Timing cached reads: 26876 MB in 1.99 seconds = 13487.78 MB/sec Timing buffered disk reads: 12694 MB in 3.00 seconds = 4230.87 MB/sec	
Partitions	Partition: ID-1: / size: 249.98 GiB used: 99.18 GiB (39.7%) fs: btrfs dev: /dev/dm-0 ID-2: /boot size: 1.9 GiB used: 816.6 MiB (42.0%) fs: ext4 dev: /dev/nvme0n1p2 ID-3: /boot/efi size: 2 GiB used: 38.9 MiB (1.9%) fs: vfat dev: /dev/nvme0n1p1 ID-4: /home size: 249.98 GiB used: 99.18 GiB (39.7%) fs: btrfs dev: /dev/dm-0 ID-5: /var/log size: 249.98 GiB used: 99.18 GiB (39.7%) fs: btrfs dev: /dev/dm-0	Drive Benchmark	fio	75.2 19300 1182 303000 Average: 646 MB/s	
			dbench	Average: 2310.04 MB/s	
Battery	Battery:			Test: Read: Average: 8749.905068 MB/s Test: Write: Average: 62057.059385 MB/s Test: Create Files: Average: 18.803995 us Per Event	

battery	ID-1: BAT0 charge: 87 Wh (96%) condition: 90.7/90 Wh (100.8%)	System Benchmark	osbench	Test: Create Threads: Average: 11.229515 us Per Event Test: Launch Programs: Average: 41.909218 us Per Event Test: Create Processes: Average: 35.67934 us Per Event Test: Memory Allocations: Average: 73.136091 Ns Per Event

Completion time: Wed 25 Mar 2026 6:26:42

Full Log

Updating and loading repositories:
Allowed part of fb-runtime prod repo 100% | 764.0 B/s | 421.0 B | 00m01s
Facebook packages for any Fedora relea 100% | 760.0 B/s | 419.0 B | 00m01s
Repositories loaded.
Package "phoronix-test-suite-10.8.4-10.fc43.noarch" is already installed.
Package "glmark2-2023.01^20250221gitcebbb63-2.fc43.x86_64" is already installed.
Package "sysbench-1.0.20-19.fc43.x86_64" is already installed.
Package "memtester-4.7.1-2.fc43.x86_64" is already installed.
Package "iozone-3.506-5.fc43.x86_64" is already installed.
Package "hdparm-9.65-9.fc43.x86_64" is already installed.
Package "stress-1.0.7-7.fc43.x86_64" is already installed.
Package "inxi-3.3.40-1.fc43.noarch" is already installed.

Nothing to do.

These are the default configuration options for when running the Phoronix Test Suite in a batch mode (i.e. running phoronix-test-suite batch-benchmark universe). Running in a batch mode is designed to be as autonomous as possible, except for where you'd like any end-user interaction.

The complete log of all tests performed will be included in the HTML report but is also written here: /root/hardware_validation.log

Manual testing

Additional testing performed besides the automated tests with the ISO above are documented below. These tests are usually performed by a QA engineer on an provisioned device, either provisioned ith the ISO in this repository or the Vanilla ISO officially provided by the Fedora Project.

Recommended Tools

- GPU Tests
 - [glmark2](#)
 - [Blender](#)
 - [Unigine](#)
- CPU tests
 - [GTKStressTesting](#)
 - [Geekbench](#)
- RAM test
 - [memtester](#) (sudo dnf install memtester)
 - [sysbench](#) (sudo dnf install sysbench)
- battery test
 - [GNOME power statistics](#) (sudo dnf install gnome-power-manager)
- Drive test
 - [KDiskMark](#) (flatpak install flathub io.github.jonmagon.kdiskmark)
- Device specs tool
 - [inxi](#) (sudo dnf install inxi)

Additional tooling to consider

- Hardware probe (hw-probe):https://fedoraproject.org/wiki/Hardware_probe
- Example output:<https://linux-hardware.org/?probe=273526bab2>

Tests to perform

Hardware tests:

- GPU tests

- [glmark2](#)
 - glmark2-wayland –fullscreen
 - Capture score output and compare with others online
- [Blender](#)
 - Ensure output matches overall GPU score from others online
- [Unigine](#)
 - Ensure test runs at ultra graphics settings with expected score
 - No visible screen tearing
- CPU tests:
 - [GTKStressTesting](#) (flatpak install flathub com.leinardi.gst)
 - System can maintain stability under 100% load for 30 mins
 - [Geekbench](#)
 - Performance scores match the hardware expectations
- Memory:
 - [memtester](#)
 - Example command: `sudo memtester 16G 3`
 - [sysbench](#)
 - Example command: `sysbench memory run`
- Disk: [KDiskMark](#) (flatpak install flathub io.github.jonmagon.kdiskmark)
 - SSD benchmark matches hardware expectations
 - Ensure CoW is disabled
- battery (if testing a laptop device): GNOME power statistics
 - For expected power consumption do the following:
 - Set power mode to Power Saver
 - Screen brightness to 50%
 - Caffeine to keep device awake
 - Run CPU stress test above to keep CPU usage at 15-20% (2-4 workers should suffice)
- TPM2.0 security chip
 - Ensure secure boot and 3rd party microsoft certs are enabled in the BIOS
 - Complete the installation with the hwqual ISO which secure boot is on
 - The installer will run through configuring akmods and enroll a self-signed certificate with MOK key enrollment
 - Enroll the drive LUKS encryption to the TPM. You can use one of the helper scripts from this repo: <https://github.com/angelothodorakis/fedora-helper-scripts>
- External ports - USB ports - charging
- Webcam: Camera app should be enough
- Microphone: <https://zoom.us/test>
- Audio Input: System settings Microphone input gauge or zoom test call.
- Audio Output: System setting audio out test or zoom test call.
- WiFi: system settings - Wi-Fi - connect to networks
- Bluetooth: system settings - Bluetooth - Connect to a bluetooth device
- Keyboard
 - During preboot (BIOS), LUKS and OS
- Mouse/Trackpad
 - During preboot (BIOS), LUKS and OS

Software tests:

- Vanilla OS installation (Workstation and/or Fedora KDE)
 - Any issues from the out of the box experience
 - Driver support
 - DNF update - LVFS firmware support (fwupdmgmr update)
- Suspend/sleep - resume
- System settings functionality
 - Function keys
 - Change screen brightness
 - Volume control
 - Turn WiFi/Bluetooth on/off

(Optional) Final go/no-go test:

- Spend a week using the device as the daily driver

This test is usually a time sensitive one and relies on the tester's discretion. It is recommended to perform since there is no better way to get a feel of a device User Experience by using it as your daily driver for at least a week.

Results

Results are captured in a spreadsheet using a provided template.

Additionally, we create and link the following 2 reports to the template so we have the exact specs of the devices qualified:

- output of `sudo inxi -e`
- output of `sudo -E hw-probe -all -upload` like here: <https://linux-hardware.org/?probe=a98b6568d5>

TODO

- Results publishing page with tester's report.
- Report template